CODEFORCES
Sponsored by Telegram

AmPieSamPie | Logout

HOME TOP CATALOG CONTESTS GYM PROBLEMSET GROUPS RATING EDU API CALENDAR HELP

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

C. A Beautiful Sorted List

time limit per test: 1 second
memory limit per test: 1024 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Alice and Bob are two friends. Alice has a list of length ***N*** in **non-decreasing** order, and Bob has a list of length ***M***, also in **non-decreasing** order.

Now, they want to combine their lists into a single non-decreasing list of length **N+M**. However, they are not very good at algorithms, so they asked for your help.

Since you are a computer science student, your task is to write an efficient algorithm to merge the two given lists into one non-decreasing list. **Solve the problem in O(N+M)**.

Input

The first line contains an integer ***N*** $(1 \leq N \leq 10^6)$, denoting the length of Alice's list.

The second line contains N space-separated integers representing Alice's list.

The third line contains an integer ***M*** $(1 \leq M \leq 10^6)$, denoting the length of Bob's list.

The fourth line contains M space-separated integers representing Bob's list.

All the numbers given in the input will fit within a **32-bit signed integer**. It is guaranteed that the given lists will be in **non-decreasing** order.

Output

You have to make a sorted list in **non-decreasing** order from the given lists and show the output.

Examples

input	Copy
4 1 3 5 7 4 2 2 4 8	

output	Copy
1 2 2 3 4 5 7 8	

input	Copy
3 2 10 12 6 3 4 6 7 8 9	

output	Copy
2 3 4 6 7 8 9 10 12	

input	Copy
5 1 2 3 4 5	

CSE221 Assignment 02 Fall 2026
Contest is running
3 days
Contestant

→ Submit?
Language: Java 21 64bit
Choose file: Choose File No file chosen
Success!
CLOUDFLARE
Privacy • Help
Submit

```
2
10 12
```

output Copy

```
1 2 3 4 5 10 12
```

input Copy

```
4
1 2 12 13
3
10 15 18
```

output Copy

```
1 2 10 12 13 15 18
```

input Copy

```
8
1 2 3 8 8 10 12 14
9
1 1 4 5 6 8 13 15 16
```

output Copy

```
1 1 1 2 3 4 5 6 8 8 8 10 12 13 14 15 16
```

Codeforces (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Oct/29/2026 09:16:55^{UTC+6} (f1).
Desktop version, switch to mobile version.
Privacy Policy | Terms and Conditions

Supported by

